

DIFFERENTIAL AND INTEGRAL
CALCULUS 1M — 104018
WINTER 2021

Exam A

January 15, 2022

Student ID Number:

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Instructions

1. **Duration: 180 minutes.**
2. This exam has **6** questions for a total of **100** points.
3. Write your answers clearly, completely and concisely in the space provided. **Quote** any results that you use. **No credit will be given to answers which are not justified.**
4. Use of calculators, personal dictionaries, electronic devices, reference materials, personal notes or any other extra material is not allowed. You will be provided with a dictionary (by demand) on the examination class (by one of the proctors).

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1. Let a and b be real constants and consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = \begin{cases} \frac{bx^2 - ax}{x-2} & \text{if } x \leq -1; \\ \frac{a(x+2)}{4-x^2} & \text{if } -1 < x \leq 1; \\ \frac{x^2 + ax + 1}{x+1} & \text{if } x > 1. \end{cases}$$

- (a) **(5 points)** Find the values of a and b for which f is continuous on $\mathbb{R}$.
- (b) **(10 points)** For these values of a and b , determine all the asymptotes of f .

Answer

(a) f IS CONTINUOUS ON $(-\infty, -1) \cup (-1, 1) \cup (1, +\infty)$ BECAUSE IT IS GIVEN BY RATIONAL EXPRESSIONS WHERE THE DENOMINATOR IS NOT 0. SO IT REMAINS TO GUARANTEE THAT f IS CONTINUOUS AT $x = -1$ AND $x = 1$.

$$\underline{x = -1} \quad \lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^-} \frac{bx^2 - ax}{x-2} = \frac{b+a}{-3} = f(-1)$$

$$\lim_{x \rightarrow -1^+} f(x) = \lim_{x \rightarrow -1^+} \frac{a(x+2)}{4-x^2} = \frac{a}{3}$$

THUS, f IS CONTINUOUS AT $x = -1$ IF AND ONLY IF

$$\frac{b+a}{-3} = \frac{a}{3} \Leftrightarrow 2a + b = 0$$

$$\underline{x = 1} \quad \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \frac{a(x+2)}{4-x^2} = a = f(1)$$

$$\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{x^2 + ax + 1}{x+1} = \frac{a+2}{2}$$

THUS, f IS CONTINUOUS AT $x=1$ IF AND ONLY IF

$$a = \frac{a+2}{2} \Leftrightarrow a = 2$$

THEREFORE, f IS CONTINUOUS IF AND ONLY IF $a=2$

$$\text{AND } 2a + b = 0 \Leftrightarrow b = -2a = -4$$

(b) FOR $a=2$, $b=-4$, f IS CONTINUOUS ON $\mathbb{R}$,
SO THERE ARE NO VERTICAL ASYMPTOTES. FOR THE
HORIZONTAL AND OBLIQUE/SLANT ASYMPTOTES, WE STUDY
THE BEHAVIOUR OF f WHEN $x \rightarrow \pm\infty$.

$$\begin{aligned} \text{FOR } x \leq -1, \quad f(x) &= \frac{-4x^2 - 2x}{x-2} = \frac{(x-2)(-4x-10) - 20}{x-2} \\ &= -4x - 10 - \frac{20}{x-2} \end{aligned}$$

SO $y = -4x - 10$ IS AN OBLIQUE ASYMPTOTE AT $-\infty$. IN
PARTICULAR, THERE IS NO HORIZONTAL ASYMPTOTE AT $-\infty$ (BECAUSE
 $\lim_{x \rightarrow -\infty} f(x) = +\infty$).

$$\text{FOR } x > 1, \quad f(x) = \frac{x^2 + 2x + 1}{x+1} = \frac{(x+1)^2}{x+1} = x+1.$$

SO, AS ABOVE, $y = x+1$ IS AN OBLIQUE ASYMPTOTE
AT $+\infty$ AND THERE IS NO HORIZONTAL ASYMPTOTE AT $+\infty$
(BECAUSE $\lim_{x \rightarrow +\infty} f(x) = +\infty$).

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2. **(20 points)** Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function which is twice differentiable on (a, b) . Assume that there exists $c \in (a, b)$ such that $f(a) = f(c) = f(b)$. Prove that there is at least one point $\alpha \in (a, b)$ such that $f''(\alpha) = 0$.

Answer

By ROLLE'S THEOREM (RT) APPLIED TO f ON $[a, c]$ AND ON $[c, b]$, THERE EXIST $\alpha_1 \in (a, c)$, $\alpha_2 \in (c, b)$ SUCH THAT $f'(\alpha_1) = 0 = f'(\alpha_2)$.

SINCE $(\alpha_1, \alpha_2) \subseteq (a, b)$ AND f IS TWICE DIFFERENTIABLE ON (a, b) , IT FOLLOWS THAT f' IS CONTINUOUS ON $[\alpha_1, \alpha_2]$ AND DIFFERENTIABLE ON (α_1, α_2) .

THUS, BY RT, THERE IS SOME $\alpha \in (\alpha_1, \alpha_2) \subseteq (a, b)$ SUCH THAT $f''(\alpha) = 0$.

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3. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = e^x(ax^2 + b),$$

where a and b are real constants.

- (a) **(10 points)** Under which conditions does f have at least two critical points?
- (b) **(10 points)** Assume $a = 4$ and $b = 3$. Find the increasing and decreasing intervals of f , as well as its local extrema.

Answer

(a) f IS DIFFERENTIABLE ON $\mathbb{R}$, SO THE CRITICAL POINTS ARE THE SOLUTIONS OF $f'(x) = 0$. WE HAVE

$$f'(x) = e^x(ax^2 + 2ax + b).$$

THUS, $f'(x) = 0 \Leftrightarrow ax^2 + 2ax + b = 0$.

SO f HAS AT LEAST 2 CRITICAL POINTS IF AND ONLY IF $ax^2 + 2ax + b$ HAS AT LEAST 2 SOLUTIONS. THUS, EITHER $a = b = 0$ OR

$$a \neq 0 \text{ AND } (2a)^2 - 4ab > 0.$$

SINCE $(2a)^2 - 4ab = 4a(a - b)$, WE CAN DEDUCE THAT f HAS AT LEAST 2 CRITICAL POINTS IF AND ONLY IF EITHER $a = b = 0$ OR

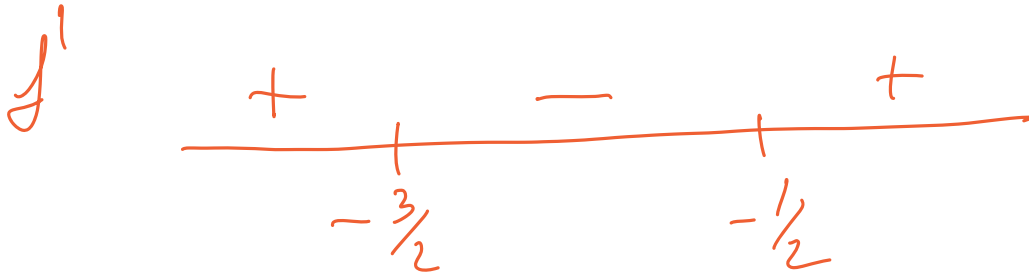
$$\boxed{(a > 0 \text{ AND } a > b)} \text{ OR } \boxed{(a < 0 \text{ AND } a < b)}.$$

(b) $a = 4$, $b = 3$. THEN,

$$f'(x) = 0 \Leftrightarrow 4x^2 + 8x + 3 = 0 \Leftrightarrow x = \frac{-8 \pm 4}{8}$$

$$\Leftrightarrow x = -\frac{3}{2} \quad \text{OR} \quad x = -\frac{1}{2}$$

THE SIGN OF f' IS GIVEN BELOW:



THUS:

f IS INCREASING ON $(-\infty, -\frac{3}{2})$ AND ON $(-\frac{1}{2}, +\infty)$

f IS DECREASING ON $(-\frac{3}{2}, -\frac{1}{2})$.

By THE FIRST DERIVATIVE TEST, f HAS A

LOCAL MAXIMUM AT $x = -\frac{3}{2}$

LOCAL MINIMUM AT $x = -\frac{1}{2}$

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4. Compute the following indefinite integrals.

(a) (10 points)

$$\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$$

(b) (5 points)

$$\int x \ln \left[\frac{1+x}{1-x} \right] dx$$

Answer

(a) LET'S MAKE THE SUBSTITUTION $u = x^{1/6}$, SO THAT
 $x = u^6$ AND $\frac{dx}{du} = 6u^5$. THEN

$$\int \frac{dx}{\sqrt{x} + \sqrt[3]{x}} = \int \frac{6u^5 du}{u^3 + u^2} = \int \frac{6u^3}{u+1} du = 6 \int \frac{u^3}{u+1} du$$

SINCE $u^3 = (u+1)(u^2 - u + 1) - 1$, WE HAVE

$$\begin{aligned} 6 \int \frac{u^3}{u+1} du &= 6 \int \left(u^2 - u + 1 - \frac{1}{u+1} \right) du \\ &= 6 \left(\frac{u^3}{3} - \frac{u^2}{2} + u - \ln(u+1) \right) + C \\ &= 2\sqrt{x} - 3\sqrt[3]{x} + 6\sqrt[6]{x} - 6\ln(\sqrt[6]{x} + 1) + C \end{aligned}$$

(b) WE USE INTEGRATION BY PARTS, WITH
ASSUME $\frac{1+x}{1-x} > 0$

$$u = \ln \left(\frac{1+x}{1-x} \right), \quad dv = x dx.$$

$$\begin{aligned} \text{THUS } du &= \frac{1-x}{1+x} \cdot \left(\frac{1+x}{1-x} \right)' dx = \frac{1-x}{1+x} \cdot \frac{1-x + (1+x)}{(1-x)^2} dx = \frac{2 dx}{(1+x)(1-x)} \\ &= \frac{2 dx}{1-x^2} \\ v &= \frac{x^2}{2} \end{aligned}$$

$$\text{SO } \int x \ln \left(\frac{1+x}{1-x} \right) dx = \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) - \int \frac{x^2 dx}{1-x^2}$$

$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + \int \frac{x^2 dx}{x^2-1} = \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + \int \frac{(x^2-1)+1}{x^2-1} dx$$

$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + \int 1 + \frac{1}{(x-1)(x+1)} dx = \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + x + \int \left(\frac{\frac{1}{2}}{x-1} - \frac{\frac{1}{2}}{x+1} \right) dx$$

$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + x + \frac{1}{2} \left(\ln |x-1| - \ln |x+1| \right) + C$$

$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + x + \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + C$$

$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + x - \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| + C$$

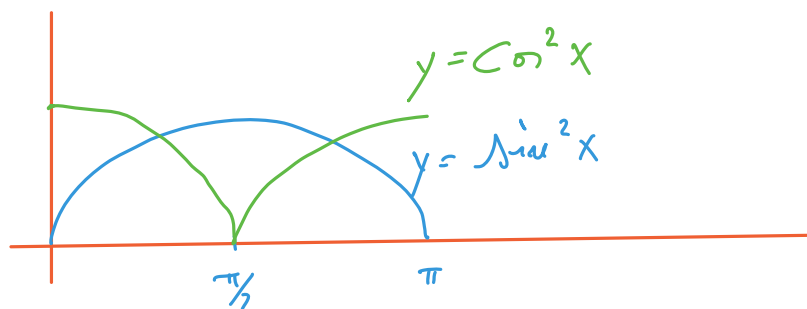
$$= \frac{x^2}{2} \ln \left(\frac{1+x}{1-x} \right) + x - \frac{1}{2} \ln \left(\frac{x+1}{1-x} \right) + C$$

$$= \frac{1}{2} (x^2-1) \ln \left(\frac{1+x}{1-x} \right) + x + C$$

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5. (15 points) Determine the minimal value of $a > 0$ such that the area of the region limited by $y = \sin^2 x$, $y = 0$, $x = 0$ and $x = a$ is equal to the area of the region limited by $y = \cos^2 x$, $y = 0$, $x = 0$ and $x = a$.

Answer



$$\int_0^a \sin^2 x \, dx = \int_0^a \frac{1 - \cos 2x}{2} \, dx = \left[\frac{x - \frac{\sin 2x}{2}}{2} \right]_0^a$$
$$\frac{1}{2} \left(a - \frac{\sin 2a}{2} \right) \quad \left(\cos^2 x = \frac{1 + \cos 2x}{2} \right)$$

So,

$$\int_0^a \cos^2 x \, dx = \int_0^a (1 - \sin^2 x) \, dx = \int_0^a 1 \, dx - \int_0^a \sin^2 x \, dx$$
$$= a - \frac{1}{2} \left(a - \frac{\sin 2a}{2} \right) = \frac{1}{2} \left(a + \frac{\sin 2a}{2} \right)$$

THEREFORE, WE WANT THE SMALLEST POSITIVE a SUCH THAT

$$\frac{1}{2} \left(a - \frac{\sin 2a}{2} \right) = \frac{1}{2} \left(a + \frac{\sin 2a}{2} \right)$$

→ thus, $\sin 2a = 0 \Leftrightarrow 2a = k\pi$, for some $k \in \mathbb{Z}$

→ THE SMALLEST SUCH $a > 0$ IS $a = \frac{\pi}{2}$.

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6. (a) (7 points) For which values of $a \in \mathbb{R}$ does the series

$$\sum_{n=1}^{\infty} \frac{a}{n(n+1)^a}$$

converge? Justify your answer.

(b) (8 points) Study the convergence of the following series:

$$\sum_{n=1}^{\infty} \frac{3 + (-1)^n}{2^{n+1}}.$$

(a) IF $a=0$ THEN $\sum_{n=1}^{\infty} \frac{a}{n(n+1)^a}$ CONVERGES.

IF $a \neq 0$, THEN

$$\sum_{n=1}^{\infty} \frac{a}{n(n+1)^a} \text{ CONVERGES } \Leftrightarrow \sum_{n=1}^{\infty} \frac{1}{n(n+1)^a} \text{ CONVERGES}$$

↓
(MULTIPLY / DIVIDE BY $a \neq 0$)

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{n(n+1)^a}}{\frac{1}{n^{a+1}}} = \lim_{n \rightarrow \infty} \frac{n^{a+1}}{n(n+1)^a} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^a = 1^a = 1.$$

So

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)^a} \text{ CONVERGES } \Leftrightarrow \sum_{n=1}^{\infty} \frac{1}{n^{a+1}} \text{ CONVERGES}$$

THE LATTER IS A p -SERIES, WITH $p = a+1$, WHICH CONVERGES IF AND ONLY IF $a+1 > 1 \Leftrightarrow a > 0$.

$$\text{So } \sum_{n=1}^{\infty} \frac{a}{n(n+1)^a} \text{ CONVERGES } \Leftrightarrow a \geq 0.$$

$$(b) \quad 0 \leq \frac{3 + (-1)^n}{2^{n+1}} \leq \frac{4}{2^{n+1}} = \frac{1}{2^{n-1}}, \quad \text{FOR ALL } n \geq 1.$$

SINCE $\sum_{n \geq 1} \frac{1}{2^{n-1}}$ IS A GEOMETRIC SERIES

WITH RATIO $R = \frac{1}{2}$ AND $|R| < 1$,

THE SERIES CONVERGES $\left(\sum_{n \geq 1} \frac{1}{2^{n-1}} = \frac{1}{1 - \frac{1}{2}} = 2 \right).$

BY THE COMPARISON TEST,

$\sum_{n \geq 1} \frac{3 + (-1)^n}{2^{n+1}}$ ALSO CONVERGES.

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